

Fourth Semester
Name of Paper: Environment and Ecology
Course No.: BCA/4/CC/19

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

1. Introduction.

Scope and importance-multidisciplinary nature of environmental studies
Ecology-concept, classification and ecological levels of organization
Ecosystem-concept, types, classification and components of an ecosystem
Structure and function of an ecosystem

2. Natural Resources

Introduction, classification of natural resources
Basics of renewable resources
Basics of non-renewable resources
Causes for depletion and conservation measures of natural resources

3. Biodiversity and Conservation

Definition, pattern and levels of biodiversity
Concept of mega-diversity and hotspot of biodiversity
Causes and consequences' of biodiversity
Conservation of biodiversity – *in-situ and Ex-situ*

4. Environment Pollution

Basic concept and types of environmental pollution
Sources and causes of environmental pollution (air, water, soil, noise)
Effects and control measures of environmental pollution (air, water, soil, noise)
Climate change-global warming, ozone layer depletion, acid rain and impact on human.
Communities and agriculture

5. Social Issues, Environment and Development

Goals, objectives and guiding principles of environmental education
Environmental movements- Chipko, Appiko, Silent Valley projects.
Roles of NGOs in environment development
Concept of sustainable development.

Suggested Readings:

1. Asthana, D.K. and Asthana, M.(2010): *A Text Book of Environmental Studies*. S. Chand and Company, New Delhi.
2. Bharucha, E.(2013): *Text Book of Environmental Studies*. University Press (India) Pvt Ltd,Hyderabad.
3. Daniel, D.C. (2014): *Environmental Science*. Jones and Bartlett Publishers, London

Fourth Semester
Name of Paper: Web Programming
Course No.: BCA/4/CC/20

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1: Basics of PHP

Introduction to PHP – what does PHP Do? – a brief history of PHP – language basics – lexical structure – data types – variables – expressions and operators – flow control statements – including code – embedding PHP in web pages.

Unit-2: Functions & Strings

Functions & Strings: Calling a function – defining a function – variable scope – function parameters – return values – variable functions – anonymous functions- User defined function, Strings: Accessing individual characters – cleaning strings – encoding and escaping – comparing strings – manipulating and searching strings – regular expression.

Unit-3: Arrays & Objects

Arrays and Objects: Indexed Vs associative arrays – identifying elements of an array – storing data in arrays – multidimensional arrays – extracting multiple values – converting between arrays and variables – traversing arrays – sorting. Objects: Creating an object – accessing properties and methods – declaring a class – introspection.

Unit-4: MySQL An Overview

Introduction – connecting to and disconnecting from the server – Entering queries – Creating and using a database – Creating and selecting a database – creating a table – loading data into a table – Retrieving information from a table – selecting all data – selecting particular rows – selecting particular columns – sorting rows – date calculations – working with NULL values – pattern matching – counting rows – using more than one tables.

Unit-5: MySQL Databases In PHP

Introduction – connecting to a MySQL database – querying the database – Retrieving and displaying the results – modifying data – deleting data, SQL injection, Backing up and restoring MySQL database managing users and privileges, Designing simple applications

Recommended Books

1. Rasmus Lerdorf, Kevin Tatroe, Bob Kaehms, Ric McGredy (2002), *Programming PHP*, O'Reilly(SPD). (Unit I,II & III)
2. Lee Babin, Nathan A. Good, Frank M. Kromann, Jon Stephens (2005), *PHP 5 Recipes, A Problem Solution Approach*, Après.(Unit IV & V)
3. Vikram Vaswani (2008), *PHP: A Beginner's Guide*, McGraw-Hill

Fourth Semester
Name of Paper: Computer Networking- I
Course No.: BCA/4/CC/21

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1: Basic Concept of Networking :

(10 L)

Introduction, **Components of network:** Computer Networking (Advantages, Types of network), **Connecting devices:** Bridges, Repeater, Switches, Routers, Modem, Gateway, Hub, Wifi-routers.

Client/Server Method of Connecting Computers, Physical structure and topology, categories of network, OSI model, TCP/IP protocol suite. **Data and signals:** Analog and digital signal, transmission impairment, performance.

Unit-2: Physical layer:

(10 L)

Transmission Media: Guided Media, Unguided Media, **Digital and analog transmission:** Digital to analog conversion (ASK, FSK, PSK), Analog to digital conversion (only PCM).

Multiplexing: FDM, WDM, and TDM. **Switching:** Circuit switch network, datagram network.

Unit-3: Data Link Layer:

(10 L)

Introduction: Error detection and correction, Linear Block code: Simple parity check code, hamming codes, CRC. **Flow and error control:** Stop and wait, Stop-and-Wait ARQ, Go-Back-N ARQ and Selective repeat ARQ. PPP (framing, transition phase only).

UNIT-4: Network and Transport Layer:

(10 L)

Network Layer: Introduction, IPv4 overview, IPv4 and IPv6 address, Network address translation, Network Management, Network Elements, ICMP. Routing algorithms- Flooding, Distance-vector routing, link state routing.

Transport layer: function, process to process delivery, UDP operation, TCP (feature, segment, connection establishment and termination).

Unit-5: Application layer and LAN's:

(10 L)

Application layer: DNS, TELNET, E-mail, FTP, WWW (only architecture), HTTP, RTP.

LAN: Ethernet (Standard, Fast and Gigabit Ethernet), IEEE 802.11 (Architecture, MAC sublayer), Bluetooth (Architecture, layer), Subnetting.

Recommended Books:

1. Behrouz A Forouzan: *Data Communication and Networking*, 4th Edition, Tata Mc Graw Hill Publication (2006)
2. Andrew S. Tanenbaum: *Computer Networks*, Prentice Hall Publication, 5 editions (2010)
3. Alberto Leon-Garcia and Indra Widjaja: *Communication Network*, 2nd Edition, Tata Mc Graw Hill Publication (2003)
4. William Stalling: *Data and Computer Communication*, 10th Edition, Prentice-Hall India Publication (2013)

Fourth Semester
Name of Paper: Object Oriented Programming in C++
Course No.: BCA/4/CC/22(a)

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1:

(10 L)

Principle of OOP & Elements of C++: Basic concepts of OOP, comparison of procedural programming and OOP, advantages of OOP. **Definitions:** Class, objects, concept of inheritance and encapsulation, operator overloading, dynamic binding. Basic program construction. Tokens and identifiers. Variables and constants. Data types. Console I/O. Control statements.

Unit-2:

(10 L)

Function, Classes and Objects: Declaration of functions, calling functions, function definition, passing arguments and returning values. Return statement, types of functions, passing and returning structure variables. Overloaded functions, inline functions, default arguments, returning by reference; Declaration of classes and objects, class definition, declaration of members, object as date time, object as function arguments, arrays of objects, returning objects from function, structures and classes.

Unit-3:

(10L)

Constructors and Destructor, Operator Overloading: Basic constructor, parameterized constructors, constructor with default arguments, dynamic initialization of objects, use of copy constructor, dynamic constructors. Destructors, constraints on constructors and destructors. Overloading unary operators, overloading binary operators, rules for overloading operators

Unit-4

(10 L)

Derived Classes & Inheritance, Pointers, Virtual Function: Derived classes and base class, Class hierarchies. Inheritance, different types of inheritance, Virtual base class, Constructors in derived class. Addresses and pointers, pointers and functions, pointers and array, pointers and string, memory management using new and delete operators. Pointers to pointers, pointers to objects. Virtual Function and polymorphism, friend function.

Unit-5:

(10 L)

Streams, Exception Handling, **Class Libraries:** Stream class hierarchy, header files, ios flag, File streams, classes for file stream operations, opening and closing a file, working with single file, working with multiple files, detecting end of file, file modes, file pointers, error handling during file operations, command line arguments. Exception handling-try, catch, throw statement. Templates- Generic functions, generic class, template functions. STL, components of STL: containers, algorithms and iterators, categories of containers.

Recommended Books:

1. E Balagurusamy: *Object-Oriented Programming with C++*, 2nd Edition, Tata McGraw Hill (2001).
2. K.V. Venugopal, R.Kumar and T.Tavaishankar: *Mastering C++*, 1st Edition, Tata McGraw Hill. (1997)

Fourth Semester
Name of Paper: Object Oriented Programming in Java
Course No.: BCA/4/CC/22(b)

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit 1: Introduction to Java

(10 L)

- 1.1 Features of java
- 1.2 JDK Environment & tools like (java,javac, appletviewer, javadoc, jdb)
- 1.3 OOPs Concepts: Class, Abstraction, Encapsulation, Inheritance, Polymorphism
- 1.4 Difference between C++ and JAVA
- 1.5 Structure of java program
- 1.6 Data types, Variables, Operators ,Keywords ,Naming Convention
- 1.7 Decision Making (if, switch), Looping (for, while)
- 1.8 Array: Creating an array, Types of Array- One Dimensional arrays
- Two-Dimensional array
- 1.9 String
 - Arrays, Methods.
 - StringBuffer class

Unit 2: Classes and Objects

(10 L)

- 2.1 Creating Classes and objects
- 2.2 Memory allocation for objects
- 2.3 Constructor
- 2.4 Implementation of Inheritance: Simple, Multilevel,
- 2.5 Abstract classes and methods
- 2.6 Implementation of Polymorphism
- 2.7 Method Overloading, Method Overriding
- 2.8 Nested and Inner classes.
- 2.9 Modifiers and Access Control

Unit 3: Errors, Exceptions and File Handling:

(10 L)

Errors and Exceptions

- 3.1 Introduction, Types of Errors
- 3.2 Exceptions, Types of Exception
- 3.3 Syntax of Exception Handling Code,
Using try catch and multiple catch, Nested try, throw, throws and finally
- 3.4 Creating user defined Exceptions

File Handling

- 3.5 Stream: ByteStream Classes; CharacterStream Classes
- 3.6 File IO basics
- 3.7 File operations
 - Creating file
 - Reading file (character, byte)
 - Writing file (character, byte)

Unit 4: Packages and Interfaces File Handling

(10 L)

- 4.1 Packages
 - Packages Concept

Importing Packages
Types of Packages
Creating user defined packages

4.2 Wrapper Class

Interfaces

4.3 Defining Interfaces
4.4 Extending Interfaces
4.5 Implementing Interfaces
4.6 Nested Interfaces
4.7 Accessing Interface variables

Unit 5: Applet, AWT and Swing Programming

(10 L)

Applet

5.1 Introduction
5.2 Types of applet
5.3 Applet Life cycle
- Creating applet
- Applet tag

5.4 Applet Classes

- Color
- Graphics
- Font

AWT

5.5 Components and container used in AWT
5.6 Layout managers
5.7 Listeners and Adapter classes
5.8 Event Delegation model

Swing

5.9 Introduction to Swing Component and Container Classes

Recommended Books:

1. E Balugurusamy: *Programming with JAVA, A primer*, McGraw Hill Education
2. Herbert Schildt: *The Complete Reference, –JAVA*. 9th Edn.

Fourth Semester
Name of Paper: Web Programming using PHP Lab
Course No.: BCA/4/CC/23

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **3**
(0- 0- 3)

- A. Record Book (10 marks)**
- B. Viva Voce (15 marks)**
- C. Practical (50 marks)**

List of Experiments

1. XAMPP or WAMP installation, php syntax, comments, echo and print, Variables, PHP constants, Data types and operators.
2. PHP control Structures: if, else-if, Nested If-else, Switch Case.
3. PHP Loop: For loop, Foreach loop, While loop and Do-while loop.
4. PHP forms: GET, POST, FORM methods. Extract()
5. File Handling: File READ and WRITE, Create Image gallery in PHP.
6. File upload and File Download
7. State Management: PHP Cookies, PHP Sessions
8. Array in PHP: Index, Associated, Nested Array. Array Functions.
9. PHP Captcha: Captcha and Dynamic image Generation.
10. PHP Function and Error Handling: Local and Global and static variables, parameterized function, recursive, function parameters. Fatal Error and Warning Error
11. PHP String , String manipulation and PHP Numeric, PHP Mathematics, PHP Object Oriented
12. PHP and MYSQL (DATABASE, CONNECT, CREATE,SELECT, INSERT, WHERE, DELETE, JOINT, ORDER-BY)
13. Create a PHP page to load dynamic text using AJAX get/post
14. Create a simple webpage using PHP.
15. Creating simple applications using PHP and MySQL
16. PHP user defined functions

***Note:** The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary*

Fourth Semester
Name of Paper: C++ Programming Lab
Course No.: BCA/4/CC/24(a)

Marks Scale: **100** marks (End Sem. Exam: 75+Internal: 25)

Credit: **3**
(0- 0- 3)

A. Record Book	(10 marks)
B. Viva Voce	(15 marks)
C. Practical	(50 marks)

List of Programs to be experiment in the Lab:

1. Write a program to define a class and to create an object of it.
2. Write a program to declare static data members and static member functions of a class.
3. Write a program to create an array of objects of a class.
4. Write a program to demonstrate how to pass an object as argument to a function and return the same object.
5. Write a program to demonstrate inline function concept.
6. Write a program to demonstrate friend function concept.
7. Write a program to demonstrate virtual function concept.
8. Write a program to demonstrate working of function overloading.
9. Write a program to demonstrate working of constructor in a class.
10. Write a program to demonstrate constructor overloading in a class.
11. Write a program to demonstrate concept of copy constructor of a class.
12. Write a program to demonstrate concept of destructor of a class.
13. Write a program to demonstrate how to overload unary operator.
14. Write a program to demonstrate how to overload binary operator.
15. Write a program to demonstrate conversion from one class to another class.
16. Write a program to demonstrate single inheritance concept.
17. Write a program to demonstrate multilevel inheritance concept.
18. Write a program to demonstrate multiple inheritance concepts.
19. Write a program to demonstrate importance of virtual base class.
20. Write a program to demonstrate working of Abstract class.
21. Write a program to demonstrate working of constructors in a derived class.
22. Write a program to demonstrate “this” pointer uses.
23. Write a program to demonstrate basic file operations like open, read, write and close.
24. Write a program to demonstrate how to update a file using random access.
25. Write a program to demonstrate exception handling mechanism.
26. Write a program to demonstrate bubble sort using template function.

***Note:** The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary*

Fourth Semester
Name of Paper: Java Programming Lab
Course No.: BCA/4/CC/24(b)

Marks Scale: **100** marks (End Sem. Exam: 75 + Internal: 25)

Credit: **3**
(0 - 0- 3)

- A. Record Book (10 marks)**
B. Viva Voce (15 marks)
C. Practical (50 marks)

Guidelines:

- The output of the programs should be neatly formatted.
- The source code should be indented.
- The programs need to be interactive.
- Data validations can be done wherever applicable.
- Include comments to improve the readability of the program.

Use meaningful variable names

- 1) Demonstrating the use of methods of Math class.
- 2) Programs to implement the methods of String class
- 3) Write a program to demonstrate constructors and constructors overloading.
- 4) Write a program to demonstrate method overloading.
- 5) Write a program to demonstrate method overriding.
- 6) Write a program to demonstrate abstract class.
- 7) Write a program to demonstrate interfaces.
- 8) Write a program to demonstrate inheritance.
- 9) Write a program to demonstrate **super** and **this** keywords.
- 10) Write a program to demonstrate static variables and methods.
- 11) Write a program to demonstrate Exceptions Handling
- 12) Write a program to demonstrate FileInputStream and FileOutputStream Classes.
- 13) Write a program to demonstrate the creation and accessing of packages.
- 14) Write a program to demonstrate the creation and accessing of applets.
- 15) Write a program to demonstrate Mouse and Keyboard events in applets.
- 16) Write a program to demonstrate the creation of a frame.
- 17) Write a program to demonstrate Labels and Buttons with proper events.
- 18) Write a program to demonstrate Checkboxes with proper events.
- 19) Write a program to demonstrate CheckBoxGroups with proper events.
- 20) Write a program to demonstrate Lists and TextFields with proper events.
- 21) Write a program to demonstrate ScrollBars with proper events.
- 22) Write a program to demonstrate MenuBars and Menus.
- 23) Write a program to demonstrate Dialog boxes.
- 24) Write a program to demonstrate Calculator.

Note: The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary